Newman-Penrose Formalism Calculations

The Rotating Charged C Metric

Metric

$$ds^2 = \left(-\frac{A^2B^2{K_t}^2a^2y^4\mathrm{Gx}\left(x\right) - 2\,A^2B^2{K_t}^2a^2y^2\mathrm{Gx}\left(x\right) + A^2B^2{K_t}^2a^2\mathrm{Gx}\left(x\right) + \left(A^4B^2{K_t}^2a^4x^4 + 2\,A^2B^2{K_t}^2a^2x^2 + B^2{K_t}^2\right)\mathrm{Gy}\left(y\right)}{2\,A^4a^2x^3y^3 - A^4a^2x^2y^4 - A^2x^2 + 2\,A^2xy - (A^4a^2x^4 + A^2)y^2}\right)\mathrm{d}\tau \otimes \mathrm{d}\tau + \left(-\frac{A^2B^2K_\phi K_ta^3y^4\mathrm{Gx}\left(x\right) - B^2K_\phi K_ta\mathrm{Gx}\left(x\right) - \left(A^2B^2K_\phi K_ta^3 - B^2K_\phi K_ta\right)y^2\mathrm{Gx}\left(x\right) - \left(A^2B^2K_\phi K_ta^3x^4 - B^2K_\phi K_ta - \left(A^2B^2K_\phi K_ta^3 - B^2K_\phi K_ta\right)x^2\right)\mathrm{Gy}\left(y\right)}{2\,A^3a^2x^3y^3 - A^3a^2x^2y^4 - Ax^2 + 2\,Axy - (A^3a^2x^4 + A)y^2}\right)\mathrm{d}\tau \otimes \mathrm{d}\phi + \left(-\frac{A^2B^2a^2x^2y^2 + B^2}{(A^2x^2 - 2\,A^2xy + A^2y^2)\mathrm{Gy}\left(y\right)}\right)\mathrm{d}y \otimes \mathrm{d}y + \left(\frac{A^2B^2a^2x^2y^2 + B^2}{A^2x^2\mathrm{Gx}\left(x\right) - 2\,A^2xy\mathrm{Gx}\left(x\right) + A^2y^2\mathrm{Gx}\left(x\right)}\right)\mathrm{d}x \otimes \mathrm{d}x + \left(-\frac{A^2B^2K_\phi K_ta^3y^4\mathrm{Gx}\left(x\right) - B^2K_\phi K_ta\mathrm{Gx}\left(x\right) - \left(A^2B^2K_\phi K_ta^3 - B^2K_\phi K_ta\right)y^2\mathrm{Gx}\left(x\right) - \left(A^2B^2K_\phi K_ta^3x^4 - B^2K_\phi K_ta - \left(A^2B^2K_\phi K_ta^3 - B^2K_\phi K_ta\right)x^2\right)\mathrm{Gy}\left(y\right)}{2\,A^3a^2x^3y^3 - A^3a^2x^2y^4 - Ax^2 + 2\,Axy - (A^3a^2x^4 + A)y^2}\right)\mathrm{d}\phi \otimes \mathrm{d}\tau + \left(-\frac{A^4B^2{K_\phi}^2a^4y^4\mathrm{Gx}\left(x\right) + 2\,A^2B^2{K_\phi}^2a^2y^2\mathrm{Gx}\left(x\right) + B^2{K_\phi}^2\mathrm{Gx}\left(x\right) + \left(A^2B^2{K_\phi}^2a^2x^4 - 2\,A^2B^2{K_\phi}^2a^2x^2 + A^2B^2{K_\phi}^2a^2\right)\mathrm{Gy}\left(y\right)}{2\,A^4a^2x^3y^3 - A^4a^2x^2y^4 - A^2x^2 + 2\,A^2xy - (A^4a^2x^4 + A^2)y^2}\right)\mathrm{d}\phi \otimes \mathrm{d}\phi$$

Coordinates

$$(x^\mu) = (\tau, y, x, \phi)$$

Null Tetrad

Covariant components

$$l = \left(-\frac{\sqrt{2}ABK_ta^2x^2\sqrt{-\frac{\mathrm{Gy}(y)}{A^2a^2x^2y^2+1}}}{2\left(x-y\right)} - \frac{\sqrt{2}BK_t\sqrt{-\frac{\mathrm{Gy}(y)}{A^2a^2x^2y^2+1}}}{2\,A(x-y)}\right)\,d\tau + \left(\frac{\sqrt{2}B}{2\,A(x-y)\sqrt{-\frac{\mathrm{Gy}(y)}{A^2a^2x^2y^2+1}}}\right)\,dy + \left(\frac{\sqrt{2}BK_\phi ax^2\sqrt{-\frac{\mathrm{Gy}(y)}{A^2a^2x^2y^2+1}}}{2\left(x-y\right)} - \frac{\sqrt{2}BK_\phi a\sqrt{-\frac{\mathrm{Gy}(y)}{A^2a^2x^2y^2+1}}}{2\left(x-y\right)}\right)\,d\phi$$

$$n = \left(-\frac{\sqrt{2}ABK_ta^2x^2\sqrt{-\frac{\mathrm{Gy}(y)}{A^2a^2x^2y^2+1}}}{2\left(x-y\right)} - \frac{\sqrt{2}BK_t\sqrt{-\frac{\mathrm{Gy}(y)}{A^2a^2x^2y^2+1}}}{2\,A(x-y)}\right)\,d\tau + \left(-\frac{\sqrt{2}B}{2\,A(x-y)\sqrt{-\frac{\mathrm{Gy}(y)}{A^2a^2x^2y^2+1}}}\right)\,dy + \left(\frac{\sqrt{2}BK_\phi ax^2\sqrt{-\frac{\mathrm{Gy}(y)}{A^2a^2x^2y^2+1}}}{2\left(x-y\right)} - \frac{\sqrt{2}BK_\phi a\sqrt{-\frac{\mathrm{Gy}(y)}{A^2a^2x^2y^2+1}}}{2\left(x-y\right)}\right)\,d\phi$$

$$m = \left(\frac{i\sqrt{2}BK_t a y^2 \sqrt{\frac{\mathrm{Gx}(x)}{A^2a^2x^2y^2+1}}}{2(x-y)} - \frac{i\sqrt{2}BK_t a \sqrt{\frac{\mathrm{Gx}(x)}{A^2a^2x^2y^2+1}}}{2(x-y)} \right) d\tau + \left(-\frac{\sqrt{2}B}{2A(x-y)\sqrt{\frac{\mathrm{Gx}(x)}{A^2a^2x^2y^2+1}}} \right) dx + \left(\frac{i\sqrt{2}ABK_\phi a^2 y^2 \sqrt{\frac{\mathrm{Gx}(x)}{A^2a^2x^2y^2+1}}}{2(x-y)} + \frac{i\sqrt{2}BK_\phi \sqrt{\frac{\mathrm{Gx}(x)}{A^2a^2x^2y^2+1}}}{2A(x-y)} \right) d\phi$$

$$\overline{m} = \left(-\frac{i\sqrt{2}BK_t a y^2 \sqrt{\frac{\mathrm{Gx}(x)}{A^2a^2x^2y^2+1}}}{2(x-y)} + \frac{i\sqrt{2}BK_t a \sqrt{\frac{\mathrm{Gx}(x)}{A^2a^2x^2y^2+1}}}{2(x-y)} \right) d\tau + \left(-\frac{\sqrt{2}B}{2A(x-y)\sqrt{\frac{\mathrm{Gx}(x)}{A^2a^2x^2y^2+1}}} \right) dx + \left(-\frac{i\sqrt{2}ABK_\phi a^2 y^2 \sqrt{\frac{\mathrm{Gx}(x)}{A^2a^2x^2y^2+1}}}{2(x-y)} - \frac{i\sqrt{2}BK_\phi \sqrt{\frac{\mathrm{Gx}(x)}{A^2a^2x^2y^2+1}}}{2A(x-y)} \right) d\phi$$

Contravariant components

$$\begin{aligned} l = \Bigg(& \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^3a^2xy^2}{2(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t)\sqrt{\mathrm{Gy}(y)}} - \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^3a^2y^3}{2(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t)\sqrt{\mathrm{Gy}(y)}} \\ & + \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}Ax}{2(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t)\sqrt{\mathrm{Gy}(y)}} - \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}Ay}{2(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t)\sqrt{\mathrm{Gy}(y)}} \Bigg) d\tau \\ & + \left(-\frac{\sqrt{-A^2a^2x^2y^2-1}Ax\sqrt{\mathrm{Gy}(y)}}{\sqrt{2}A^2Ba^2x^2y^2+\sqrt{2}B} + \frac{\sqrt{-A^2a^2x^2y^2-1}Ay\sqrt{\mathrm{Gy}(y)}}{\sqrt{2}A^2Ba^2x^2y^2+\sqrt{2}B} \right) dy \\ & + \Bigg(-\frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^2axy^2}{2(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi)\sqrt{\mathrm{Gy}(y)}} + \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^2ay^3}{2(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi)\sqrt{\mathrm{Gy}(y)}} \\ & + \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^2ax}{2(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi)\sqrt{\mathrm{Gy}(y)}} - \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^2ay}{2(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi)\sqrt{\mathrm{Gy}(y)}} \Bigg) d\phi \end{aligned}$$

$$\begin{aligned} n = \Bigg(& \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^3a^2xy^2}{2(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t)\sqrt{\mathrm{Gy}(y)}} - \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^3a^2y^3}{2(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t)\sqrt{\mathrm{Gy}(y)}} \\ & + \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}Ax}{2(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t)\sqrt{\mathrm{Gy}(y)}} - \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}Ay}{2(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t)\sqrt{\mathrm{Gy}(y)}} \Bigg) d\tau \\ & + \left(\frac{\sqrt{-A^2a^2x^2y^2-1}Ax\sqrt{\mathrm{Gy}(y)}}{\sqrt{2}A^2Ba^2x^2y^2+\sqrt{2}B} - \frac{\sqrt{-A^2a^2x^2y^2-1}Ay\sqrt{\mathrm{Gy}(y)}}{\sqrt{2}A^2Ba^2x^2y^2+\sqrt{2}B} \right) dy \\ & + \Bigg(-\frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^2axy^2}{2(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi)\sqrt{\mathrm{Gy}(y)}} + \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^2ay^3}{2(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi)\sqrt{\mathrm{Gy}(y)}} \\ & + \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^2ax}{2(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi)\sqrt{\mathrm{Gy}(y)}} - \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^2ay}{2(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi)\sqrt{\mathrm{Gy}(y)}} \Bigg) d\phi \end{aligned}$$

$$m = \left(\frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^2ax^3}{2\left(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t\right)\sqrt{\mathrm{Gx}\left(x\right)}} - \frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^2ax^2y}{2\left(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t\right)\sqrt{\mathrm{Gx}\left(x\right)}} - \frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^2ax}{2\left(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t\right)\sqrt{\mathrm{Gx}\left(x\right)}} + \frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^2ay}{2\left(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t\right)\sqrt{\mathrm{Gx}\left(x\right)}} \right) d\tau + \left(-\frac{\sqrt{A^2a^2x^2y^2+1}Ax\sqrt{\mathrm{Gx}\left(x\right)}}{\sqrt{2}A^2Ba^2x^2y^2+\sqrt{2}B} + \frac{\sqrt{A^2a^2x^2y^2+1}Ay\sqrt{\mathrm{Gx}\left(x\right)}}{\sqrt{2}A^2Ba^2x^2y^2+\sqrt{2}B} \right) dx + \left(\frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^3a^2x^3}{2\left(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi\right)\sqrt{\mathrm{Gx}\left(x\right)}} - \frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^3a^2x^2y}{2\left(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi\right)\sqrt{\mathrm{Gx}\left(x\right)}} + \frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}Ax}{2\left(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi\right)\sqrt{\mathrm{Gx}\left(x\right)}} - \frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}Ay}{2\left(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi\right)\sqrt{\mathrm{Gx}\left(x\right)}} \right) d\phi$$

$$\overline{m} = \left(-\frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^2ax^3}{2\left(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t\right)\sqrt{\mathrm{Gx}\left(x\right)}} + \frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^2ax^2y}{2\left(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t\right)\sqrt{\mathrm{Gx}\left(x\right)}} + \frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^2ax}{2\left(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t\right)\sqrt{\mathrm{Gx}\left(x\right)}} - \frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^2ay}{2\left(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t\right)\sqrt{\mathrm{Gx}\left(x\right)}} \right) d\tau + \left(-\frac{\sqrt{A^2a^2x^2y^2+1}Ax\sqrt{\mathrm{Gx}\left(x\right)}}{\sqrt{2}A^2Ba^2x^2y^2+\sqrt{2}B} + \frac{\sqrt{A^2a^2x^2y^2+1}Ay\sqrt{\mathrm{Gx}\left(x\right)}}{\sqrt{2}A^2Ba^2x^2y^2+\sqrt{2}B} \right) dx + \left(-\frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^3a^2x^3}{2\left(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi\right)\sqrt{\mathrm{Gx}\left(x\right)}} + \frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^3a^2x^2y}{2\left(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi\right)\sqrt{\mathrm{Gx}\left(x\right)}} - \frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}Ax}{2\left(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi\right)\sqrt{\mathrm{Gx}\left(x\right)}} + \frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}Ay}{2\left(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi\right)\sqrt{\mathrm{Gx}\left(x\right)}} \right) d\phi$$

Directional Derivatives

$$D = \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^3a^2xy^2}{2\left(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t\right)\sqrt{\mathrm{Gy}\left(y\right)}} - \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^3a^2y^3}{2\left(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t\right)\sqrt{\mathrm{Gy}\left(y\right)}}$$
$$+ \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}Ax}{2\left(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t\right)\sqrt{\mathrm{Gy}\left(y\right)}} - \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}Ay}{2\left(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t\right)\sqrt{\mathrm{Gy}\left(y\right)}}\,\partial_\tau$$
$$- \frac{\sqrt{-A^2a^2x^2y^2-1}Ax\sqrt{\mathrm{Gy}\left(y\right)}}{\sqrt{2}A^2Ba^2x^2y^2+\sqrt{2}B} + \frac{\sqrt{-A^2a^2x^2y^2-1}Ay\sqrt{\mathrm{Gy}\left(y\right)}}{\sqrt{2}A^2Ba^2x^2y^2+\sqrt{2}B}\,\partial_y - \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^2axy^2}{2\left(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi\right)\sqrt{\mathrm{Gy}\left(y\right)}}$$
$$+ \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^2ay^3}{2\left(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi\right)\sqrt{\mathrm{Gy}\left(y\right)}} + \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^2ax}{2\left(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi\right)\sqrt{\mathrm{Gy}\left(y\right)}}$$
$$- \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^2ay}{2\left(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi\right)\sqrt{\mathrm{Gy}\left(y\right)}}\,\partial_\phi$$

$$\Delta = \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^3a^2xy^2}{2\left(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t\right)\sqrt{\mathrm{Gy}\left(y\right)}} - \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^3a^2y^3}{2\left(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t\right)\sqrt{\mathrm{Gy}\left(y\right)}}$$
$$+ \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}Ax}{2\left(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t\right)\sqrt{\mathrm{Gy}\left(y\right)}} - \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}Ay}{2\left(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t\right)\sqrt{\mathrm{Gy}\left(y\right)}}\,\partial_\tau$$
$$+ \frac{\sqrt{-A^2a^2x^2y^2-1}Ax\sqrt{\mathrm{Gy}\left(y\right)}}{\sqrt{2}A^2Ba^2x^2y^2+\sqrt{2}B} - \frac{\sqrt{-A^2a^2x^2y^2-1}Ay\sqrt{\mathrm{Gy}\left(y\right)}}{\sqrt{2}A^2Ba^2x^2y^2+\sqrt{2}B}\,\partial_y - \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^2axy^2}{2\left(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi\right)\sqrt{\mathrm{Gy}\left(y\right)}}$$
$$+ \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^2ay^3}{2\left(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi\right)\sqrt{\mathrm{Gy}\left(y\right)}} + \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^2ax}{2\left(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi\right)\sqrt{\mathrm{Gy}\left(y\right)}}$$
$$- \frac{\sqrt{2}\sqrt{-A^2a^2x^2y^2-1}A^2ay}{2\left(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi\right)\sqrt{\mathrm{Gy}\left(y\right)}}\,\partial_\phi$$

$$\delta = \frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^2ax^3}{2\left(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t\right)\sqrt{\mathrm{Gx}\left(x\right)}} - \frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^2ax^2y}{2\left(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t\right)\sqrt{\mathrm{Gx}\left(x\right)}}$$
$$- \frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^2ax}{2\left(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t\right)\sqrt{\mathrm{Gx}\left(x\right)}} + \frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^2ay}{2\left(A^4BK_ta^4x^2y^2+A^2BK_ta^2x^2y^2+A^2BK_ta^2+BK_t\right)\sqrt{\mathrm{Gx}\left(x\right)}}\,\partial_\tau - \frac{\sqrt{A^2a^2x^2y^2+1}Ax\sqrt{\mathrm{Gx}\left(x\right)}}{\sqrt{2}A^2Ba^2x^2y^2+\sqrt{2}B}$$
$$+ \frac{\sqrt{A^2a^2x^2y^2+1}Ay\sqrt{\mathrm{Gx}\left(x\right)}}{\sqrt{2}A^2Ba^2x^2y^2+\sqrt{2}B}\,\partial_x + \frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^3a^2x^3}{2\left(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi\right)\sqrt{\mathrm{Gx}\left(x\right)}} - \frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^3a^2x^2y}{2\left(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi\right)\sqrt{\mathrm{Gx}\left(x\right)}}$$
$$+ \frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}Ax}{2\left(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi\right)\sqrt{\mathrm{Gx}\left(x\right)}} - \frac{i\sqrt{2}\sqrt{A^2a^2x^2y^2+1}Ay}{2\left(A^4BK_\phi a^4x^2y^2+A^2BK_\phi a^2x^2y^2+A^2BK_\phi a^2+BK_\phi\right)\sqrt{\mathrm{Gx}\left(x\right)}}\,\partial_\phi$$

$$\begin{aligned}\overline{\delta} = & -\frac{i\sqrt{2}\sqrt{A^{2}a^{2}x^{2}y^{2}+1}A^{2}ax^{3}}{2\left(A^{4}BK_{t}a^{4}x^{2}y^{2}+A^{2}BK_{t}a^{2}x^{2}y^{2}+A^{2}BK_{t}a^{2}+BK_{t}\right)\sqrt{\mathrm{Gx}\left(x\right)}}+\frac{i\sqrt{2}\sqrt{A^{2}a^{2}x^{2}y^{2}+1}A^{2}ax^{2}y}{2\left(A^{4}BK_{t}a^{4}x^{2}y^{2}+A^{2}BK_{t}a^{2}x^{2}y^{2}+A^{2}BK_{t}a^{2}+BK_{t}\right)\sqrt{\mathrm{Gx}\left(x\right)}}\\ & +\frac{i\sqrt{2}\sqrt{A^{2}a^{2}x^{2}y^{2}+1}A^{2}ax}{2\left(A^{4}BK_{t}a^{4}x^{2}y^{2}+A^{2}BK_{t}a^{2}x^{2}y^{2}+A^{2}BK_{t}a^{2}+BK_{t}\right)\sqrt{\mathrm{Gx}\left(x\right)}}-\frac{i\sqrt{2}\sqrt{A^{2}a^{2}x^{2}y^{2}+1}A^{2}ay}{2\left(A^{4}BK_{t}a^{4}x^{2}y^{2}+A^{2}BK_{t}a^{2}x^{2}y^{2}+A^{2}BK_{t}a^{2}+BK_{t}\right)\sqrt{\mathrm{Gx}\left(x\right)}}\,\partial_{\tau}-\frac{\sqrt{A^{2}a^{2}x^{2}y^{2}+1}Ax\sqrt{\mathrm{Gx}\left(x\right)}}{\sqrt{2}A^{2}Ba^{2}x^{2}y^{2}+\sqrt{2}B}\\ & +\frac{\sqrt{A^{2}a^{2}x^{2}y^{2}+1}Ay\sqrt{\mathrm{Gx}\left(x\right)}}{\sqrt{2}A^{2}Ba^{2}x^{2}y^{2}+\sqrt{2}B}\,\partial_{x}-\frac{i\sqrt{2}\sqrt{A^{2}a^{2}x^{2}y^{2}+1}A^{3}a^{2}x^{3}}{2\left(A^{4}BK_{\phi}a^{4}x^{2}y^{2}+A^{2}BK_{\phi}a^{2}x^{2}y^{2}+A^{2}BK_{\phi}a^{2}+BK_{\phi}\right)\sqrt{\mathrm{Gx}\left(x\right)}}+\frac{i\sqrt{2}\sqrt{A^{2}a^{2}x^{2}y^{2}+1}A^{3}a^{2}x^{2}y}{2\left(A^{4}BK_{\phi}a^{4}x^{2}y^{2}+A^{2}BK_{\phi}a^{2}x^{2}y^{2}+A^{2}BK_{\phi}a^{2}+BK_{\phi}\right)\sqrt{\mathrm{Gx}\left(x\right)}}\\ & -\frac{i\sqrt{2}\sqrt{A^{2}a^{2}x^{2}y^{2}+1}Ax}{2\left(A^{4}BK_{\phi}a^{4}x^{2}y^{2}+A^{2}BK_{\phi}a^{2}x^{2}y^{2}+A^{2}BK_{\phi}a^{2}+BK_{\phi}\right)\sqrt{\mathrm{Gx}\left(x\right)}}+\frac{i\sqrt{2}\sqrt{A^{2}a^{2}x^{2}y^{2}+1}Ay}{2\left(A^{4}BK_{\phi}a^{4}x^{2}y^{2}+A^{2}BK_{\phi}a^{2}x^{2}y^{2}+A^{2}BK_{\phi}a^{2}+BK_{\phi}\right)\sqrt{\mathrm{Gx}\left(x\right)}}\,\partial_{\phi}\end{aligned}$$

Spin Coefficients

$$\begin{aligned}\rho = & \frac{\sqrt{-A^{2}a^{2}x^{2}y^{2}-1}A^{3}a^{2}x^{3}y\sqrt{\mathrm{Gy}\left(y\right)}}{\sqrt{2}A^{4}Ba^{4}x^{4}y^{4}+2\sqrt{2}A^{2}Ba^{2}x^{2}y^{2}+\sqrt{2}B}-\frac{i\sqrt{-A^{2}a^{2}x^{2}y^{2}-1}A^{2}ax^{2}\sqrt{\mathrm{Gy}\left(y\right)}}{\sqrt{2}A^{4}Ba^{4}x^{4}y^{4}+2\sqrt{2}A^{2}Ba^{2}x^{2}y^{2}+\sqrt{2}B}\\ & +\frac{i\sqrt{-A^{2}a^{2}x^{2}y^{2}-1}A^{2}axy\sqrt{\mathrm{Gy}\left(y\right)}}{\sqrt{2}A^{4}Ba^{4}x^{4}y^{4}+2\sqrt{2}A^{2}Ba^{2}x^{2}y^{2}+\sqrt{2}B}+\frac{\sqrt{-A^{2}a^{2}x^{2}y^{2}-1}A\sqrt{\mathrm{Gy}\left(y\right)}}{\sqrt{2}A^{4}Ba^{4}x^{4}y^{4}+2\sqrt{2}A^{2}Ba^{2}x^{2}y^{2}+\sqrt{2}B}\end{aligned}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\begin{aligned}\mu = & \frac{\sqrt{-A^{2}a^{2}x^{2}y^{2}-1}A^{3}a^{2}x^{3}y\sqrt{\mathrm{Gy}\left(y\right)}}{\sqrt{2}A^{4}Ba^{4}x^{4}y^{4}+2\sqrt{2}A^{2}Ba^{2}x^{2}y^{2}+\sqrt{2}B}-\frac{i\sqrt{-A^{2}a^{2}x^{2}y^{2}-1}A^{2}ax^{2}\sqrt{\mathrm{Gy}\left(y\right)}}{\sqrt{2}A^{4}Ba^{4}x^{4}y^{4}+2\sqrt{2}A^{2}Ba^{2}x^{2}y^{2}+\sqrt{2}B}\\ & +\frac{i\sqrt{-A^{2}a^{2}x^{2}y^{2}-1}A^{2}axy\sqrt{\mathrm{Gy}\left(y\right)}}{\sqrt{2}A^{4}Ba^{4}x^{4}y^{4}+2\sqrt{2}A^{2}Ba^{2}x^{2}y^{2}+\sqrt{2}B}+\frac{\sqrt{-A^{2}a^{2}x^{2}y^{2}-1}A\sqrt{\mathrm{Gy}\left(y\right)}}{\sqrt{2}A^{4}Ba^{4}x^{4}y^{4}+2\sqrt{2}A^{2}Ba^{2}x^{2}y^{2}+\sqrt{2}B}\end{aligned}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = \frac{\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^3a^2x^3y^2\frac{\partial}{\partial x}\mathrm{Gx}\left(x\right)}{8\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)\sqrt{\mathrm{Gx}\left(x\right)}} - \frac{\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^3a^2x^2y^3\frac{\partial}{\partial x}\mathrm{Gx}\left(x\right)}{8\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)\sqrt{\mathrm{Gx}\left(x\right)}} - \frac{\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^3a^2x^2y^2\sqrt{\mathrm{Gx}\left(x\right)}}{2\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)}$$
$$+ \frac{\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^3a^2xy^3\sqrt{\mathrm{Gx}\left(x\right)}}{4\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)} + \frac{i\,\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^2axy\sqrt{\mathrm{Gx}\left(x\right)}}{4\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)} - \frac{i\,\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^2ay^2\sqrt{\mathrm{Gx}\left(x\right)}}{4\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)}$$
$$+ \frac{\sqrt{2}\sqrt{A^2a^2x^2y^2+1}Ax\frac{\partial}{\partial x}\mathrm{Gx}\left(x\right)}{8\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)\sqrt{\mathrm{Gx}\left(x\right)}} - \frac{\sqrt{2}\sqrt{A^2a^2x^2y^2+1}Ay\frac{\partial}{\partial x}\mathrm{Gx}\left(x\right)}{8\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)\sqrt{\mathrm{Gx}\left(x\right)}} - \frac{\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A\sqrt{\mathrm{Gx}\left(x\right)}}{4\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)}$$

$$\beta = -\frac{\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^3a^2x^3y^2\frac{\partial}{\partial x}\mathrm{Gx}\left(x\right)}{8\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)\sqrt{\mathrm{Gx}\left(x\right)}} + \frac{\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^3a^2x^2y^3\frac{\partial}{\partial x}\mathrm{Gx}\left(x\right)}{8\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)\sqrt{\mathrm{Gx}\left(x\right)}} + \frac{\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^3a^2x^2y^2\sqrt{\mathrm{Gx}\left(x\right)}}{2\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)}$$
$$- \frac{\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^3a^2xy^3\sqrt{\mathrm{Gx}\left(x\right)}}{4\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)} - \frac{i\,\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^2axy\sqrt{\mathrm{Gx}\left(x\right)}}{4\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)} + \frac{i\,\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^2ay^2\sqrt{\mathrm{Gx}\left(x\right)}}{4\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)}$$
$$- \frac{\sqrt{2}\sqrt{A^2a^2x^2y^2+1}Ax\frac{\partial}{\partial x}\mathrm{Gx}\left(x\right)}{8\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)\sqrt{\mathrm{Gx}\left(x\right)}} + \frac{\sqrt{2}\sqrt{A^2a^2x^2y^2+1}Ay\frac{\partial}{\partial x}\mathrm{Gx}\left(x\right)}{8\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)\sqrt{\mathrm{Gx}\left(x\right)}} + \frac{\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A\sqrt{\mathrm{Gx}\left(x\right)}}{4\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)}$$

$$\tau = \frac{i\,\sqrt{-A^2a^2x^2y^2-1}A^3a^2xy^3\sqrt{\mathrm{Gx}\left(x\right)}}{\sqrt{2}A^4Ba^4x^4y^4+2\,\sqrt{2}A^2Ba^2x^2y^2+\sqrt{2}B} - \frac{\sqrt{-A^2a^2x^2y^2-1}A^2axy\sqrt{\mathrm{Gx}\left(x\right)}}{\sqrt{2}A^4Ba^4x^4y^4+2\,\sqrt{2}A^2Ba^2x^2y^2+\sqrt{2}B}$$
$$+ \frac{\sqrt{-A^2a^2x^2y^2-1}A^2ay^2\sqrt{\mathrm{Gx}\left(x\right)}}{\sqrt{2}A^4Ba^4x^4y^4+2\,\sqrt{2}A^2Ba^2x^2y^2+\sqrt{2}B} + \frac{i\,\sqrt{-A^2a^2x^2y^2-1}A\sqrt{\mathrm{Gx}\left(x\right)}}{\sqrt{2}A^4Ba^4x^4y^4+2\,\sqrt{2}A^2Ba^2x^2y^2+\sqrt{2}B}$$

$$\pi = -\frac{i\,\sqrt{-A^2a^2x^2y^2-1}A^3a^2xy^3\sqrt{\mathrm{Gx}\left(x\right)}}{\sqrt{2}A^4Ba^4x^4y^4+2\,\sqrt{2}A^2Ba^2x^2y^2+\sqrt{2}B} + \frac{\sqrt{-A^2a^2x^2y^2-1}A^2axy\sqrt{\mathrm{Gx}\left(x\right)}}{\sqrt{2}A^4Ba^4x^4y^4+2\,\sqrt{2}A^2Ba^2x^2y^2+\sqrt{2}B}$$
$$- \frac{\sqrt{-A^2a^2x^2y^2-1}A^2ay^2\sqrt{\mathrm{Gx}\left(x\right)}}{\sqrt{2}A^4Ba^4x^4y^4+2\,\sqrt{2}A^2Ba^2x^2y^2+\sqrt{2}B} - \frac{i\,\sqrt{-A^2a^2x^2y^2-1}A\sqrt{\mathrm{Gx}\left(x\right)}}{\sqrt{2}A^4Ba^4x^4y^4+2\,\sqrt{2}A^2Ba^2x^2y^2+\sqrt{2}B}$$

$$\epsilon = -\frac{i\,\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^3a^2x^3y^2\frac{\partial}{\partial y}\mathrm{Gy}\left(y\right)}{8\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)\sqrt{\mathrm{Gy}\left(y\right)}} + \frac{i\,\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^3a^2x^2y^3\frac{\partial}{\partial y}\mathrm{Gy}\left(y\right)}{8\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)\sqrt{\mathrm{Gy}\left(y\right)}} + \frac{i\,\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^3a^2x^3y\sqrt{\mathrm{Gy}\left(y\right)}}{4\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)}$$
$$- \frac{i\,\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^3a^2x^2y^2\sqrt{\mathrm{Gy}\left(y\right)}}{2\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)} + \frac{\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^2ax^2\sqrt{\mathrm{Gy}\left(y\right)}}{4\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)} - \frac{\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A^2axy\sqrt{\mathrm{Gy}\left(y\right)}}{4\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)}$$
$$- \frac{i\,\sqrt{2}\sqrt{A^2a^2x^2y^2+1}Ax\frac{\partial}{\partial y}\mathrm{Gy}\left(y\right)}{8\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)\sqrt{\mathrm{Gy}\left(y\right)}} + \frac{i\,\sqrt{2}\sqrt{A^2a^2x^2y^2+1}Ay\frac{\partial}{\partial y}\mathrm{Gy}\left(y\right)}{8\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)\sqrt{\mathrm{Gy}\left(y\right)}} - \frac{i\,\sqrt{2}\sqrt{A^2a^2x^2y^2+1}A\sqrt{\mathrm{Gy}\left(y\right)}}{4\left(A^4Ba^4x^4y^4+2\,A^2Ba^2x^2y^2+B\right)}$$

$$
\begin{aligned}
\gamma = & -\frac{i\sqrt{2}\sqrt{A^{2}a^{2}x^{2}y^{2}+1}A^{3}a^{2}x^{3}y^{2}\frac{\partial}{\partial y}\mathrm{Gy}\left(y\right)}{8\left(A^{4}Ba^{4}x^{4}y^{4}+2\,A^{2}Ba^{2}x^{2}y^{2}+B\right)\sqrt{\mathrm{Gy}\left(y\right)}}+\frac{i\sqrt{2}\sqrt{A^{2}a^{2}x^{2}y^{2}+1}A^{3}a^{2}x^{2}y^{3}\frac{\partial}{\partial y}\mathrm{Gy}\left(y\right)}{8\left(A^{4}Ba^{4}x^{4}y^{4}+2\,A^{2}Ba^{2}x^{2}y^{2}+B\right)\sqrt{\mathrm{Gy}\left(y\right)}}+\frac{i\sqrt{2}\sqrt{A^{2}a^{2}x^{2}y^{2}+1}A^{3}a^{2}x^{3}y\sqrt{\mathrm{Gy}\left(y\right)}}{4\left(A^{4}Ba^{4}x^{4}y^{4}+2\,A^{2}Ba^{2}x^{2}y^{2}+B\right)} \\
& -\frac{i\sqrt{2}\sqrt{A^{2}a^{2}x^{2}y^{2}+1}A^{3}a^{2}x^{2}y^{2}\sqrt{\mathrm{Gy}\left(y\right)}}{2\left(A^{4}Ba^{4}x^{4}y^{4}+2\,A^{2}Ba^{2}x^{2}y^{2}+B\right)}+\frac{\sqrt{2}\sqrt{A^{2}a^{2}x^{2}y^{2}+1}A^{2}ax^{2}\sqrt{\mathrm{Gy}\left(y\right)}}{4\left(A^{4}Ba^{4}x^{4}y^{4}+2\,A^{2}Ba^{2}x^{2}y^{2}+B\right)}-\frac{\sqrt{2}\sqrt{A^{2}a^{2}x^{2}y^{2}+1}A^{2}axy\sqrt{\mathrm{Gy}\left(y\right)}}{4\left(A^{4}Ba^{4}x^{4}y^{4}+2\,A^{2}Ba^{2}x^{2}y^{2}+B\right)} \\
& -\frac{i\sqrt{2}\sqrt{A^{2}a^{2}x^{2}y^{2}+1}Ax\frac{\partial}{\partial y}\mathrm{Gy}\left(y\right)}{8\left(A^{4}Ba^{4}x^{4}y^{4}+2\,A^{2}Ba^{2}x^{2}y^{2}+B\right)\sqrt{\mathrm{Gy}\left(y\right)}}+\frac{i\sqrt{2}\sqrt{A^{2}a^{2}x^{2}y^{2}+1}Ay\frac{\partial}{\partial y}\mathrm{Gy}\left(y\right)}{8\left(A^{4}Ba^{4}x^{4}y^{4}+2\,A^{2}Ba^{2}x^{2}y^{2}+B\right)\sqrt{\mathrm{Gy}\left(y\right)}}-\frac{i\sqrt{2}\sqrt{A^{2}a^{2}x^{2}y^{2}+1}A\sqrt{\mathrm{Gy}\left(y\right)}}{4\left(A^{4}Ba^{4}x^{4}y^{4}+2\,A^{2}Ba^{2}x^{2}y^{2}+B\right)}
\end{aligned}
$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$
\begin{aligned}
\Phi_{11} = & -\frac{A^{4}a^{2}x^{4}y^{2}\frac{\partial^{2}}{(\partial x)^{2}}\mathrm{Gx}\left(x\right)}{8\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)}+\frac{A^{4}a^{2}x^{3}y^{3}\frac{\partial^{2}}{(\partial x)^{2}}\mathrm{Gx}\left(x\right)}{4\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)}-\frac{A^{4}a^{2}x^{2}y^{4}\frac{\partial^{2}}{(\partial x)^{2}}\mathrm{Gx}\left(x\right)}{8\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)}-\frac{A^{4}a^{2}x^{4}y^{2}\frac{\partial^{2}}{(\partial y)^{2}}\mathrm{Gy}\left(y\right)}{8\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)} \\
& +\frac{A^{4}a^{2}x^{3}y^{3}\frac{\partial^{2}}{(\partial y)^{2}}\mathrm{Gy}\left(y\right)}{4\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)}-\frac{A^{4}a^{2}x^{2}y^{4}\frac{\partial^{2}}{(\partial y)^{2}}\mathrm{Gy}\left(y\right)}{8\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)}+\frac{3\,A^{4}a^{2}x^{3}y^{2}\frac{\partial}{\partial x}\mathrm{Gx}\left(x\right)}{4\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)}-\frac{5\,A^{4}a^{2}x^{2}y^{3}\frac{\partial}{\partial x}\mathrm{Gx}\left(x\right)}{4\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)} \\
& +\frac{A^{4}a^{2}xy^{4}\frac{\partial}{\partial x}\mathrm{Gx}\left(x\right)}{2\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)}+\frac{A^{4}a^{2}x^{4}y\frac{\partial}{\partial y}\mathrm{Gy}\left(y\right)}{2\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)}-\frac{5\,A^{4}a^{2}x^{3}y^{2}\frac{\partial}{\partial y}\mathrm{Gy}\left(y\right)}{4\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)}+\frac{3\,A^{4}a^{2}x^{2}y^{3}\frac{\partial}{\partial y}\mathrm{Gy}\left(y\right)}{4\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)} \\
& -\frac{3\,A^{4}a^{2}x^{2}y^{2}\mathrm{Gx}\left(x\right)}{2\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)}+\frac{2\,A^{4}a^{2}xy^{3}\mathrm{Gx}\left(x\right)}{A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}}-\frac{A^{4}a^{2}y^{4}\mathrm{Gx}\left(x\right)}{2\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)}-\frac{A^{4}a^{2}x^{4}\mathrm{Gy}\left(y\right)}{2\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)} \\
& +\frac{2\,A^{4}a^{2}x^{3}y\mathrm{Gy}\left(y\right)}{A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}}-\frac{3\,A^{4}a^{2}x^{2}y^{2}\mathrm{Gy}\left(y\right)}{2\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)}-\frac{A^{2}x^{2}\frac{\partial^{2}}{(\partial x)^{2}}\mathrm{Gx}\left(x\right)}{8\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)}+\frac{A^{2}xy\frac{\partial^{2}}{(\partial x)^{2}}\mathrm{Gx}\left(x\right)}{4\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)} \\
& -\frac{A^{2}y^{2}\frac{\partial^{2}}{(\partial x)^{2}}\mathrm{Gx}\left(x\right)}{8\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)}-\frac{A^{2}x^{2}\frac{\partial^{2}}{(\partial y)^{2}}\mathrm{Gy}\left(y\right)}{8\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)}+\frac{A^{2}xy\frac{\partial^{2}}{(\partial y)^{2}}\mathrm{Gy}\left(y\right)}{4\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)}-\frac{A^{2}y^{2}\frac{\partial^{2}}{(\partial y)^{2}}\mathrm{Gy}\left(y\right)}{8\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)} \\
& +\frac{A^{2}x\frac{\partial}{\partial x}\mathrm{Gx}\left(x\right)}{4\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)}-\frac{A^{2}y\frac{\partial}{\partial x}\mathrm{Gx}\left(x\right)}{4\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)}-\frac{A^{2}x\frac{\partial}{\partial y}\mathrm{Gy}\left(y\right)}{4\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)}+\frac{A^{2}y\frac{\partial}{\partial y}\mathrm{Gy}\left(y\right)}{4\left(A^{4}B^{2}a^{4}x^{4}y^{4}+2\,A^{2}B^{2}a^{2}x^{2}y^{2}+B^{2}\right)}
\end{aligned}
$$

$\Phi_{12} = 0$

$\Phi_{20} = 0$

$\Phi_{21} = 0$

$\Phi_{22} = 0$

$$\Lambda = -\frac{A^2 x^2 \frac{\partial^2}{(\partial x)^2} \mathrm{Gx}\left(x\right)}{24\left(A^2 B^2 a^2 x^2 y^2 + B^2\right)} + \frac{A^2 x y \frac{\partial^2}{(\partial x)^2} \mathrm{Gx}\left(x\right)}{12\left(A^2 B^2 a^2 x^2 y^2 + B^2\right)} - \frac{A^2 y^2 \frac{\partial^2}{(\partial x)^2} \mathrm{Gx}\left(x\right)}{24\left(A^2 B^2 a^2 x^2 y^2 + B^2\right)} + \frac{A^2 x^2 \frac{\partial^2}{(\partial y)^2} \mathrm{Gy}\left(y\right)}{24\left(A^2 B^2 a^2 x^2 y^2 + B^2\right)} - \frac{A^2 x y \frac{\partial^2}{(\partial y)^2} \mathrm{Gy}\left(y\right)}{12\left(A^2 B^2 a^2 x^2 y^2 + B^2\right)} + \frac{A^2 y^2 \frac{\partial^2}{(\partial y)^2} \mathrm{Gy}\left(y\right)}{24\left(A^2 B^2 a^2 x^2 y^2 + B^2\right)} + \frac{A^2 x \frac{\partial}{\partial x} \mathrm{Gx}\left(x\right)}{4\left(A^2 B^2 a^2 x^2 y^2 + B^2\right)} - \frac{A^2 y \frac{\partial}{\partial x} \mathrm{Gx}\left(x\right)}{4\left(A^2 B^2 a^2 x^2 y^2 + B^2\right)} + \frac{A^2 x \frac{\partial}{\partial y} \mathrm{Gy}\left(y\right)}{4\left(A^2 B^2 a^2 x^2 y^2 + B^2\right)} - \frac{A^2 y \frac{\partial}{\partial y} \mathrm{Gy}\left(y\right)}{4\left(A^2 B^2 a^2 x^2 y^2 + B^2\right)} - \frac{A^2 \mathrm{Gx}\left(x\right)}{2\left(A^2 B^2 a^2 x^2 y^2 + B^2\right)} + \frac{A^2 \mathrm{Gy}\left(y\right)}{2\left(A^2 B^2 a^2 x^2 y^2 + B^2\right)}$$

Weyl Tensor Components

$\Psi_0 = 0$

$\Psi_1 = 0$

$$
\begin{aligned}
\Psi_2 = {} & \frac{A^6 a^4 x^6 y^4 \frac{\partial^2}{(\partial x)^2} \mathrm{Gx}\,(x)}{12\,(A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2)} - \frac{A^6 a^4 x^5 y^5 \frac{\partial^2}{(\partial x)^2} \mathrm{Gx}\,(x)}{6\,(A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2)} \\
& + \frac{A^6 a^4 x^4 y^6 \frac{\partial^2}{(\partial x)^2} \mathrm{Gx}\,(x)}{12\,(A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2)} - \frac{A^6 a^4 x^6 y^4 \frac{\partial^2}{(\partial y)^2} \mathrm{Gy}\,(y)}{12\,(A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2)} \\
& + \frac{A^6 a^4 x^5 y^5 \frac{\partial^2}{(\partial y)^2} \mathrm{Gy}\,(y)}{6\,(A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2)} - \frac{A^6 a^4 x^4 y^6 \frac{\partial^2}{(\partial y)^2} \mathrm{Gy}\,(y)}{12\,(A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2)} \\
& - \frac{A^6 a^4 x^5 y^4 \frac{\partial}{\partial x} \mathrm{Gx}\,(x)}{2\,(A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2)} + \frac{A^6 a^4 x^4 y^5 \frac{\partial}{\partial x} \mathrm{Gx}\,(x)}{A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2} \\
& - \frac{A^6 a^4 x^3 y^6 \frac{\partial}{\partial x} \mathrm{Gx}\,(x)}{2\,(A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2)} + \frac{A^6 a^4 x^6 y^3 \frac{\partial}{\partial y} \mathrm{Gy}\,(y)}{2\,(A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2)} \\
& - \frac{A^6 a^4 x^5 y^4 \frac{\partial}{\partial y} \mathrm{Gy}\,(y)}{A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2} + \frac{A^6 a^4 x^4 y^5 \frac{\partial}{\partial y} \mathrm{Gy}\,(y)}{2\,(A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2)} \\
& + \frac{A^6 a^4 x^4 y^4 \mathrm{Gx}\,(x)}{A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2} - \frac{2\,A^6 a^4 x^3 y^5 \mathrm{Gx}\,(x)}{A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2} + \frac{A^6 a^4 x^2 y^6 \mathrm{Gx}\,(x)}{A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2} \\
& - \frac{A^6 a^4 x^6 y^2 \mathrm{Gy}\,(y)}{A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2} + \frac{2\,A^6 a^4 x^5 y^3 \mathrm{Gy}\,(y)}{A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2} - \frac{A^6 a^4 x^4 y^4 \mathrm{Gy}\,(y)}{A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2} \\
& + \frac{i\,A^5 a^3 x^4 y^3 \frac{\partial}{\partial x} \mathrm{Gx}\,(x)}{2\,(A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2)} - \frac{i\,A^5 a^3 x^3 y^4 \frac{\partial}{\partial x} \mathrm{Gx}\,(x)}{A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2} \\
& + \frac{i\,A^5 a^3 x^2 y^5 \frac{\partial}{\partial x} \mathrm{Gx}\,(x)}{2\,(A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2)} - \frac{i\,A^5 a^3 x^5 y^2 \frac{\partial}{\partial y} \mathrm{Gy}\,(y)}{2\,(A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2)} \\
& + \frac{i\,A^5 a^3 x^4 y^3 \frac{\partial}{\partial y} \mathrm{Gy}\,(y)}{A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2} - \frac{i\,A^5 a^3 x^3 y^4 \frac{\partial}{\partial y} \mathrm{Gy}\,(y)}{2\,(A^6 B^2 a^6 x^6 y^6 + 3\,A^4 B^2 a^4 x^4 y^4 + 3\,A^2 B^2 a^2 x^2 y^2 + B^2)} - \dots
\end{aligned}
$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"